BEGINNER'S PROGRAMMING GUIDE

How to Solve Any Python Problem

A simple 5-step method to think like a programmer and turn real-life ideas into code.

1. READ 3 TIMES
2. PLAIN ENGLISH
3. MAP TO PYTHON
4. CODE LINE-BY-LINE
5. TEST & FIX

The Golden Rule of Coding

"Don't start by writing code. Start by writing steps in plain English."

The 5-Step Framework Overview

STEP 1

Read 3 Times

Read requirements carefully. Find action words like loop, calculate, print, or count.

STEP 2

Plain English

Write simple steps (pseudo-code) as if explaining to a 10-year-old child.

STEP 3

Map to Python

Match each English step with a Python tool (like `for` or check `if`

Code, Test & The Math Analogy

💻Steps 4 & 5: Code & Test

Step 4 (Code): Write one single line at a time. Run and test after every tiny addition.

Step 5 (Test): Run with test numbers. If output looks wrong, fix line by line!

💡Coding is Like Math

Your teacher gives you the **formula**, but you must **apply it** to solve problems.

We teach you the Python tools; practice teaches you how to apply them!

Real Example 1: Task Duration

Goal: Calculate remaining days for project tasks using a simple Python list.

Scenario 1: Applying Steps 1 to 3

- **Step 1 (Read):** We need to check tasks, calculate remaining days, print task details, and count unfinished tasks.
- **Formula Needed:** Remaining Days = `Duration × (1 - % Complete / 100)`
- **Step 2 (Plain English):** Go through tasks one by one -> calculate remaining -> print ID -> if remaining > 0, add 1 to counter.
- **Step 3 (Map to Python):** "Go through tasks" maps to `for task in tasks:` and "if remaining > 0" maps to `if remaining > 0:`.

Scenario 1: Python Code Solution

```
# List of project tasks tasks = [ {"id": "T-101",
"duration": 5.0, "pct": 100}, {"id": "T-102", "duration":
8.0, "pct": 60}, {"id": "T-103", "duration": 3.0, "pct":
0} ] unfinished = 0 for task in tasks: remaining =
task["duration"] * (1 - task["pct"] / 100) print( f"{task['id']}: Remaining {remaining} days"
                                           ) if
remaining > 0: unfinished += 1 print( f"Unfinished tasks: {unfinished}"
          )
```

Real Example 2: Live API Data

Goal: Fetch tasks dynamically from Primavera P6 using a Web API, then calculate remaining duration.

Connecting to Live Web APIs

1. Login First

Send login credentials to P6 server to get permission to access project data.

2. Fetch Tasks

Use API requests (`requests.get` sk lists directly over internet.

3. Same Logic!

Once data arrives as JSON, use the exact same calculation loop as before!

Static Data vs Web API Data

Feature	Scenario 1 (Static List)	Scenario 2 (P6 API Data)
Data Source	Hardcoded inside Python code	Live from Primavera P6 server
Setup Required	None (Ready to use)	API login & JSON response parsing
Main Calculation	Loop → Calculate → Print → Count	**Identical!** Loop → Calculate → Print
Main Purpose	Learning & testing concepts	Real-world automation & reporting

Your Turn: Practice Exercise

Problem Statement

You have a list of team resources and their assigned work hours.

1. Ask user for a resource name.
2. Find all tasks for that person.
3. Print each task and calculate total hours.

Steps to Follow

Rule #1: Do not write code right away!

- Write Step 2 (Plain English steps) first.
- Identify where you need a loop inside a loop.
- Test line-by-line!

Keep Practicing!

Stuck on a problem? Show your plain English steps first!

🎓 Python Beginner Series | Step-by-Step Problem Solving